GOVERNANCE · DOMAIN 8 - DATA GOVERNANCE

Metadata, tags, lineage and discoverability

Section 8. Making data findable, and knowing which of it to trust.

SECTION 8 of 10 WEIGHT 7% TOPIC Discoverability

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Data nobody can find has no value. Data nobody trusts has **negative** value, because somebody will use it anyway.

How the documentation frames it

- **Catalog Explorer** to browse, search and inspect what exists.
- **AI-generated comments** to describe tables and columns without waiting for humans to write them.
- **Certified and deprecated tags** to signal which assets should be trusted.
- Plus **lineage**, which answers where something came from and what depends on it.

> ON THE EXAM
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> Those first three are the documentation's own sentence about discoverability. If a question asks how to make data findable, they are the expected vocabulary.

Who can actually see them

- **Any user with BROWSE privileges can view comments.** That is what makes them a discoverability tool.
- So somebody with no access to the data at all can read what a table is for, and decide whether to request it.
- To **add or edit** a comment you must **own the object or hold MANAGE** on it.
- Comments cannot be edited on objects in a Delta Sharing catalog. In the legacy metastore, tables only.

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> This is module 27's BROWSE arriving with a purpose. Granting BROWSE broadly is only useful if there is something written down to read.

Two commands, and one exception

- **COMMENT ON** works for any securable object **except table columns**.
- For a **column**, use the comment option on the alter statement instead.
- Comments can also be added and edited directly in Catalog Explorer.
- Modifying a table comment records a properties operation in the **table history**, capturing the SQL used.

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> That last point is quietly useful. Documentation changes are versioned alongside the data, so you can see when a description changed and to what.

Markdown, and where it stops

- Comments support **basic Markdown** in Catalog Explorer. All basic syntax is supported **except images**.
- **Markdown does not render in DESCRIBE output.** There you get the raw characters.
- **Using Markdown comments has no impact on query performance.**
- So a table comment can be a genuine short document, with headings and links, rather than one sentence.

> ON THE EXAM
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> Write for Catalog Explorer, and remember that anyone reading through DESCRIBE sees the syntax rather than the formatting.

AI-generated comments

- Databricks can generate comments for you, also called **AI-generated documentation**.
- Available for **tables and table columns only**. Not for catalogs, schemas or other objects.
- Offered as **suggestions you review and apply**, rather than written silently.
- That matters at scale. Nobody is going to hand-document nine hundred tables, so the realistic choice is generated or nothing.

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> The scope limit is the examinable bit. If a question asks about documenting a catalog or a volume, generated comments are not the answer.

Certified and deprecated are a system tag

- There is a **system-governed tag** for certification status, with exactly **two values: certified and deprecated**.
- It is **not free-form**. You cannot invent a third value or define your own equivalent tag with the same meaning.
- It applies to catalogs, schemas, tables, dashboards, notebooks, apps and Genie Agents.
- The status is **displayed next to object names**, and influences how data appears in notebooks and the SQL editor.

> ON THE EXAM
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> Being system-governed is the point. Everybody's certified means the same thing, which is exactly what a free-form tag cannot guarantee.

Setting and searching certification

- Assign it from the object's menu in the interface, choosing certified, deprecated, or none.
- Or in SQL with a **set tag** statement naming the system certification status key.
- **Search for it with the certification status keyword**, so you can list every certified asset.
- You can update or remove the status at any time. Deprecating is not permanent.

> ON THE EXAM
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> Deprecated is the underused half. Marking a superseded table stops people finding it and trusting it, which is often more valuable than certifying a good one.

Keys, optional values, and governance

- A tag is an attribute with a **key and an optional value**, applicable to any securable object.
- Tags make objects **searchable and categorisable**, which is the discovery use.
- **Governed tags** control which tags may exist and **who is permitted to apply them**.
- Without that control, tagging degrades into inconsistent free text nobody can rely on.

> ON THE EXAM
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> Governed tags went GA alongside ABAC and data classification. The three are designed as one system, not three features.

Tags do double duty

- The same tag that makes a column **findable** can be what a **masking policy matches on**.
- Applying coarse-grained tags at **catalog or schema level** means they cover nested objects.
- So one tagging scheme serves discovery and protection at once, which is why the two shipped together.
- It also raises the stakes. A tag that is wrong is now both a discovery problem and a security problem.

> ON THE EXAM
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> This is the strongest argument for governed tags. When a tag drives access control, who can apply it matters as much as what it says.

Data classification

- Databricks can **automatically scan and tag sensitive data**, such as personal data, across tables.
- That closes the loop: classification finds it, tags mark it, and policies protect it.
- To manage the configuration you must be the **catalog owner**, or hold **USE CATALOG and MANAGE** on the catalog.
- The alternative is asking every team to correctly identify their own sensitive columns, forever.

ON THE EXAM

Classification, tags and policies are three parts of one pipeline: detect, mark, protect. Knowing the shape answers design questions.

What is captured, and how

- Lineage is captured **automatically at runtime**. You do not declare it or maintain it.
- Across **all languages**, and down to the **column level**.
- It includes the **notebooks, jobs and dashboards** related to the query, not only the tables.
- Visualised in **near real-time** in Catalog Explorer, and queryable from system tables.

> ON THE EXAM
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> Automatic capture is the important word. Hand-maintained lineage diagrams are wrong within weeks; this is derived from what actually ran.

The questions it answers

- **Where did this come from?** Which sources feed a number somebody is querying.
- **What breaks if I change this?** Impact analysis before altering a schema or dropping a column.
- **Where did this personal data go?** Tracing sensitive columns downstream, which is module 26's problem.
- **Model lineage** also traces a model back to the datasets it was trained and evaluated on.

> ON THE EXAM
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> Column-level lineage is what makes the third question answerable. Table-level would only tell you a table was involved somewhere.

The Insights tab

- Catalog Explorer browses and manages data, schemas, tables, permissions, owners, external locations and credentials.
- Its **Insights tab** shows the **most frequent recent queries and users** of any registered table.
- That answers a question nothing else does: **who actually uses this, and how?**
- Useful before deprecating something, and useful for finding the person who understands a table.

> ON THE EXAM
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> Lineage tells you what is connected. Insights tells you what is actually being used, which is a different and often more useful question.

A four-level namespace was announced

- At the June 2026 summit, Databricks announced a **four-level namespace**: metastore, catalog, schema, table.
- The aim is a single address for every asset across accounts, regions and clouds, with one discovery experience.
- **Cross-region governance is described as coming soon in preview.** Cross-cloud and cross-account follow.
- The current documentation still describes the **three-level** namespace, so that is the examinable answer.

> ON THE EXAM
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> Announced, not shipped. Module 27 teaches three-level because that is what the current docs and the exam describe.

What to carry into the exam

- Discoverability is **Catalog Explorer, AI-generated comments, and certified and deprecated tags**.
- **BROWSE lets people read comments** without data access. Writing them needs ownership or MANAGE.
- AI-generated comments cover **tables and columns only**. Markdown does not render in DESCRIBE.
- Certification is a **system-governed tag** with exactly two values. It is not free-form.
- **Lineage is automatic**, all languages, column level, including notebooks, jobs and dashboards.

> ON THE EXAM
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> That completes Section 8 at 7%. Section 9, debugging and deploying, is next at 10%.

> A platform team grants BROWSE on all catalogs to every account user, expecting people to start finding datasets themselves. Six weeks later, discovery has not improved and the data team still receives the same requests.
>
> Investigation shows almost no tables have descriptions.

What is missing?

A. BROWSE does not permit viewing comments; USE CATALOG is also required.

B. BROWSE lets users read comments, but there are almost none to read. Add comments, using AI generation at scale.

C. Users also need SELECT before search returns results.

D. Discovery requires the certification tag on every table before assets appear in search.

◆ PAUSE AND DECIDE

What is missing?

A BROWSE does not permit viewing comments; USE CATALOG is also required.

BROWSE specifically allows viewing comments without usage privileges.

B BROWSE lets users read comments, but there are almost none to read. Add comments, using AI generation at scale.

The two mechanisms are designed to work together. Access to documentation is useless when none exists.

C Users also need SELECT before search returns results.

That would defeat the purpose of discovery without data access.

D Discovery requires the certification tag on every table before assets appear in search.

Certification signals trust. It is not a prerequisite for appearing in search.

ON THE EXAM

Half a mechanism does nothing. BROWSE is the permission to read documentation, and somebody still has to write it.

> A team wants to mark superseded tables so colleagues stop using them. They propose creating a tag with the key status and the value old, and asking everyone to check it before querying.
>
> Several other teams have already invented their own conventions for the same thing.

What is the better approach?

A Standardise on the proposed status tag and document the convention widely.

B Use the system-governed certification status tag with the deprecated value, which displays next to object names.

C Add a comment on each table saying it is deprecated.

D Revoke SELECT on the superseded tables so nobody can use them.

◆ PAUSE AND DECIDE

What is the better approach?

A Standardise on the proposed status tag and document the convention widely.

Reinvents a system tag, and relies on convention where a governed mechanism exists.

B Use the system-governed certification status tag with the deprecated value, which displays next to object names.

System-governed, two fixed values, displayed beside object names and searchable. Built for exactly this.

C Add a comment on each table saying it is deprecated.

Only helps someone who opens the table and reads the comment.

D Revoke SELECT on the superseded tables so nobody can use them.

Breaks anything still legitimately reading them, with no warning period.

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The several competing conventions in the scenario are the argument for a system-governed tag: everybody's deprecated means the same thing.

> A team is asked to remove a column from a widely used silver table. They need to know what will break before doing it, including reports nobody has mentioned.
>
> They have no diagram of their pipeline dependencies.

What should they use?

A Column-level lineage, which is captured automatically and includes related notebooks, jobs and dashboards.

B The Insights tab, which shows every downstream table.

C The audit system table, which records every read of the column.

D Ask each team to declare their dependencies, then build a diagram.

◆ PAUSE AND DECIDE

What should they use?

A Column-level lineage, which is captured automatically and includes related notebooks, jobs and dashboards.

Exactly the impact-analysis case, and it covers dashboards and jobs rather than only tables.

B The Insights tab, which shows every downstream table.

Insights shows frequent recent queries and users, not the dependency graph.

C The audit system table, which records every read of the column.

Audit records access events. Lineage is the purpose-built view of dependency.

D Ask each team to declare their dependencies, then build a diagram.

Slow, and it captures only what people remember. Lineage is derived from what actually ran.

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Note what A includes. Table-level lineage would find dependent tables; dashboards and jobs are where the unmentioned reports live.

> An engineer writes a detailed table comment using Markdown headings and a bulleted list. It renders correctly in Catalog Explorer. A colleague running DESCRIBE TABLE EXTENDED reports that the comment is full of hash symbols and asterisks.
>
> They ask whether the comment is corrupted.

What should they be told?

A The comment is corrupted and should be rewritten as plain text.

B Markdown renders in Catalog Explorer but not in DESCRIBE output, where the raw syntax is returned.

C Markdown in comments degrades query performance and should be removed.

D DESCRIBE only supports comments under a fixed character limit.

◆ PAUSE AND DECIDE

What should they be told?

A The comment is corrupted and should be rewritten as plain text.

Nothing is corrupted. Both views are showing the same stored string.

B Markdown renders in Catalog Explorer but not in DESCRIBE output, where the raw syntax is returned.

Documented behaviour. Worth knowing when deciding how heavily to format a comment.

C Markdown in comments degrades query performance and should be removed.

The docs state explicitly that Markdown comments have no impact on query performance.

D DESCRIBE only supports comments under a fixed character limit.

Invented.

ON THE EXAM

Neither view is wrong. One renders the formatting and one returns the stored string, which is worth knowing before writing heavily formatted documentation.

> A governance programme needs sensitive columns tagged consistently across 1,200 tables so that masking policies apply automatically. New tables appear weekly, created by eight different teams.
>
> A proposal is to circulate a tagging standard and ask each team to tag their own columns.

What is the stronger design?

A The proposal is sound; the teams know their own data best.

B Use data classification to scan and tag automatically, with governed tags controlling who may apply them, and policies matching on the tags.

C Add AI-generated comments to every column describing its sensitivity.

D Grant BROWSE widely so the governance team can find sensitive columns manually.

◆ PAUSE AND DECIDE

What is the stronger design?

A The proposal is sound; the teams know their own data best.

True and insufficient. It depends on eight teams applying one standard consistently, forever.

B Use data classification to scan and tag automatically, with governed tags controlling who may apply them, and policies matching on the tags.

The detect, mark, protect pipeline these three features were designed as.

C Add AI-generated comments to every column describing its sensitivity.

Comments aid discovery. Policies match on tags, not on comments.

D Grant BROWSE widely so the governance team can find sensitive columns manually.

Manual discovery at 1,200 tables and growing, which is the problem rather than the solution.

ON THE EXAM

Classification, governed tags and policies were released together because they are one system. Recognising the pipeline shape answers the question.

> Before deprecating a legacy table, a team wants to know whether anyone is actually still using it, and who. Lineage shows three downstream tables, all of which were themselves deprecated last year.
>
> They need evidence of real usage, not just connections.

Where should they look?

A Lineage, which already showed them everything connected to the table.

B The Insights tab in Catalog Explorer, which shows the most frequent recent queries and users of the table.

C The certification status tag, which records who certified it.

D Table comments, where users typically record their usage.

◆ PAUSE AND DECIDE

Where should they look?

A Lineage, which already showed them everything connected to the table.

Connected is not used. The scenario says the connected objects are themselves dead.

B The Insights tab in Catalog Explorer, which shows the most frequent recent queries and users of the table.

Answers the usage question directly, including who is querying it and how often.

C The certification status tag, which records who certified it.

It records a status, not usage.

D Table comments, where users typically record their usage.

Comments are documentation. Nobody logs their queries there.

ON THE EXAM

Lineage and Insights answer different questions. What is connected, and what is actually being used. A dead pipeline is connected and unused.